

13.3 部分习题参考简答

1. (1) $\frac{1}{3}\pi H^3$; (2) $\frac{1}{6}$.
2. (1) $\int_{-1}^1 dx \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} dy \int_{\sqrt{x^2+y^2}}^1 f(x, y, z) dz$ 或 $\int_{-1}^1 dy \int_{-\sqrt{1-y^2}}^{\sqrt{1-y^2}} dx \int_{\sqrt{x^2+y^2}}^1 f(x, y, z) dz$
 (2) $I = \int_0^1 dx \int_0^{1-x} dy \int_0^{x^2+y^2} f(x, y, z) dz$ 或 $I = \int_0^1 dy \int_0^{1-y} dx \int_0^{x^2+y^2} f(x, y, z) dz$.
3. $\frac{\sin 1}{2}$.
4. $f(t) = \frac{1}{4\pi} (e^{4\pi|t|^3} - 1)$.
5. $\frac{4\pi}{3} f(0, 0, 0)$.
6. (1) $\frac{3}{4}\pi ab$; (2) $\frac{1}{48}$; (3) $\frac{\pi abc^2}{4}$; (4) $\frac{28\pi R^5}{15}$; (5) $\frac{(2-\sqrt{2})\pi R^5}{5}$;
 (6) $\frac{\pi^2 abc}{4}$.
7. (1) $\frac{\pi^2 abc}{4}$; (2) $\frac{32\pi}{3}$; (3) $\frac{(2-\sqrt{2})\pi}{3}(b^3 - a^3)$.
8. (1) $\frac{21\sqrt{3}}{8}$; (2) $\frac{1}{364}$; (3) $\frac{\sin 1}{6}$; (4) $\frac{2}{3}$.
9. $\frac{53}{60}$.
10. $\frac{1}{3}abc$.